

Exam Date & Time: 09-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. EXAMINATIONS - JANUARY 2024

SUBJECT: ECE 1071 / ECE-1071 - BASIC ELECTRONICS

**Answer ALL questions.
Missing data may be assumed.**

BASIC ELECTRONICS [ECE 1071]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Draw the VI characteristics of the Silicon diode. Explain the different types of breakdown mechanisms in a PN junction diode. (4)
- 1B) If a Silicon diode has a reverse saturation current of 20nA at 27°C , calculate the reverse current, forward current and dynamic resistance at 40°C when the diode is forward biased at 0.7V . (3)
- 1C) Obtain an expression for the drain current of MOSFET in terms of ' V_{GS} ', ' V_{DS} ' and the dimension of the device. Also, plot the VI characteristics of the device indicating all the salient features on it. (3)
- 2A) Explain the operation of half wave rectifier with neat circuit and waveform. Derive an equation for V_{dc} , V_{rms} and efficiency. (4)
- 2B) A full wave bridge rectifier with capacitor filter has $V_m = 8\text{V}$, $R_L = 100\Omega$, $C = 1000\mu\text{F}$ and $f = 50\text{Hz}$. Calculate the ripple factor and the DC output voltage. (3)
- 2C) A Zener regulator circuit has $V_Z = 20\text{V}$, $I_{Z\text{max}} = 60\text{mA}$, $R_S = 220\Omega$ and $R_L = 1.2\text{k}\Omega$. Calculate the range of input voltage for which output will remain constant. Draw the circuit of a Zener voltage regulator and label all the components. (3)
- 3A) i) Represent $(0111)_2$ in the following codes:
a) Gray code b) (7, 4) even parity Hamming code (4)
ii) Perform $(13.25)_{10} - (26.75)_{10}$ using 1's complement method.
- 3B) Draw the circuit diagram of an OPAMP inverting amplifier and derive the expression for an output voltage. Given $R_F = 20\text{k}\Omega$, $R_I = 4\text{k}\Omega$ and the dc supply voltage is $\pm 15\text{V}$. Calculate and plot the output voltage with respect to the input, if an input applied is $3\sin(2\pi 1000t)$ Volt. (3)
- 3C) Design an adder circuit using two OPAMP's to obtain: (3)

$$V_{out} = 0.75V_1 - 0.25V_2 - 1.75V_3. \text{ Assume } R_F = 10k\Omega.$$

- 4A) Explain the Full adder circuit with a neat circuit diagram and truth table. Also obtain $(45)_{10} - (87)_{10}$ using 2's complement method. (4)
- 4B) Design a 2 bit Ripple up counter using positive edge triggered JK flipflop. Also sketch the timing diagram for the same. (3)
- 4C) Explain the operation of positive edge triggered SR flipflop using NAND gate with a neat circuit diagram and truth table. Also Realize D flipflop using SR flipflop. (3)
- 5A) Define Frequency Modulation (FM). Given a FM signal $V_{FM}(t) = 20 \cos [2\pi 10^8 t + 5 \sin(2\pi 15000t)]$, Calculate Carrier frequency, Modulating frequency, Frequency deviation and Bandwidth using Carson's rule. (4)
- 5B) Define sampling theorem. For an analog signal, $x(t) = 3 \cos 100\pi t$. Determine the minimum sampling rate required to reconstruct the signal. (3)
- 5C) For the given binary data 1101 sketch the ASK, FSK and PSK waveforms. (3)

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